Feasibility-Gated Humanoid Motion Tracking: Separating Reference Physics from Curriculum Difficulty

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Abstract— **Adaptive curricula improve humanoid tracking by concentrating updates on difficult motion segments, but persistent policy error can also arise when a retargeted reference demands unsupported or over-limit dynamics. We call this *reference–physics misalignment* (RPM): the curriculum interprets an inadmissible reference as learnable difficulty. In a three-seed Unitree G1 campaign, peak top-1 allocation reached 87–89%, and the same kneel-and-crawl reference became a dominant attractor in every seed; during its descent, 86% of frames have no collision geometry within 6 cm of the floor, while median unsupported force is approximately one body weight. We introduce CLIMB, a feasibility-gated tracking pipeline that screens final robot-space trajectories with contact-free inverse dynamics and a torque-limited contact program, then converts admissible intervals into exact non-wrapping training units. Learning progress reallocates samples only inside this gate, which yields 1,184 hash-bound units and 368,951 legal fixed-horizon starts. On one AMASS-to-G1 pipeline, the screen flags 2,442 of 10,705 clips above a fixed 10%-of-frames threshold; on a separate filtered 4,950-clip production pairing, an independent implementation flags 7, and the implementations agree on 39 of 40 strict decisions in a stratified same-clip panel. *Phase-G status: endpoint-blind calibration in progress*. Its matched ALP-versus-uniform outcome will determine whether adaptive allocation adds value after reference feasibility is fixed.**

I. Introduction

Large retargeted motion banks have made generalist humanoid tracking a practical training recipe: map human motion to a robot, train a policy over the resulting references in massively parallel simulation, and spend additional updates on motions the policy currently fails. Systems such as PHC, MaskedMimic, H2O, ExBody, BeyondMimic, and SONIC demonstrate the scale and capability of this recipe [1], [2], [3], [4], [5], [6]. Failure, completion, or recent learning progress appears to tell the trainer where another update is most useful [5], [7], [8].

That interpretation assumes that policy error measures a solvable control problem. A retargeted trajectory may instead have no admissible contact capable of supplying its demanded base wrench, or require joint effort beyond the modeled actuator range. We call this *reference–physics misalignment* (RPM). Under RPM, the same persistent error can mean either that the controller has not learned a feasible skill or that the reference does not define one for this embodiment and scene. A policy-only curriculum cannot distinguish the two from failure alone.

We observed this ambiguity in a 100-motion Unitree G1 campaign. A BeyondMimic-style sampler concentrated on the same kneel-and-crawl clip in all three seeds. Peak top-1 mass was 0.884, 0.870, and 0.893; those campaign maxima belonged to another clip in two seeds. Clip-uniform sampling nevertheless achieved higher held-out survival in all three paired seeds. The attractor passes ordinary kinematic checks. Yet from 0.75 to 1.75 s its pelvis descends from 0.79 to 0.40 m while the nearest collision geometry remains 7–10 cm above the floor. A median 329 N of support demand, approximately the 327 N model weight, has no modeled source.

CLIMB closes this reference–policy interface in three stages (Fig. 1). First, a robot-space feasibility screen evaluates the final retargeted trajectory using the target MuJoCo model, modeled contacts, friction, and actuator ranges. Second, cause-aware routing admits supported intervals, sends scene-mismatch or repair candidates to separate paths, and quarantines residual failures. Third, an exact-support curriculum turns admitted intervals into non-wrapping fixed-horizon starts and applies outcome-based allocation only within that support. This design separates reference admissibility from learning progress while holding embodiment, reward, PPO, legal-start prior, caps, compute, seeds, and evaluator fixed.

Feasibility is a corpus–pipeline property. The screen flags 2,442/10,705 clips in one AMASS-to-`whole_body_tracking`-to-G1 pipeline and 7/4,950 in a separately filtered BONES-SEED-to-G1 production pairing used by SONIC. The implementations agree on 39/40 strict decisions in a stratified same-clip panel. Source corpus, filtering, robot file, and implementation all change between the bank-scale measurements, so they motivate measuring each pairing rather than comparing retargeters from unmatched denominators.

This paper contributes:

1) *RPM diagnosis*: a three-seed curriculum attractor localized to a robot-space interval whose demanded wrench has no admissible modeled support source;
2) *the CLIMB method*: a policy-independent contact/torque screen, cause-aware routing, and a hard feasibility gate yielding 1,184 units and 368,951 legal starts; and
3) *claim-isolating evaluation*: bank-scale and cross-implementation screen tests plus a matched ALP-versus-uniform allocation ablation on identical feasible support.

The allocation ablation is in endpoint-blind calibration; its exhaustive result status is positive, null, inconclusive, or `not_tested` under the predeclared manipulation and provenance gates.

II. Related Work

A. Motion filtering and dynamic retargeting

Policy-based curation already removes poorly tracked references. H2O retains AMASS motions that a privileged imitator can follow, and ExBody2 uses an initial policy's sequence errors to select a feasible and diverse subset [3], [9]. These scores combine reference quality with policy capability. KungfuBot instead uses a human-space center-of-mass/center-of-pressure heuristic before robot retargeting [10]. LIMMT combines target-robot motion heuristics, diversity, and complexity, with physical-score weights calibrated through repeated policy training [11]. PHUMA filters source artifacts and retargets with joint-limit, ground-contact, and anti-skating losses [12]. CLIMB builds on this line but changes the training interface: the final robot-space screen determines the support on which an adaptive allocator is permitted to act.

Optimization-based retargeters address a complementary problem. Kinodynamic Motion Retargeting uses rigid-body and contact constraints to construct viable locomotion, while Direct Dynamic Retargeting uses simulator-in-the-loop optimization without an infeasible geometric intermediate [13], [14]. AMO and SPIDER likewise generate viable references through trajectory optimization or physics-based sampling with contact guidance [15], [16]. GMR further shows that retargeting artifacts affect downstream tracking robustness [17]. An earlier analytic study estimates velocity, acceleration, and torque requirements for 40 upper-body motions, but does not test whole-body environmental support [18]. Repair can therefore be preferable to exclusion. Our narrower question is diagnostic: after retargeting, can the demanded robot-space wrench be supplied by admissible contacts within modeled actuator limits?

B. Adaptive allocation and evaluation

Prioritized experience and level replay formalize the benefit and bias of nonuniform training distributions [19], [20]. In humanoid tracking, BeyondMimic uses a failure-EMA bin sampler, while GMT and EGM reweight motions or segments using tracking outcomes [5], [7], [8]. Those systems combine allocation with curation, staging, clipping, or architectural changes. Empirical failure, completion, and tracking error identify hard examples, but do not by themselves determine whether a reference is dynamically admissible. CLIMB inserts that test before outcome-based allocation; its matched comparison then changes only allocation over a shared set of legal trials.

Robot-learning comparisons are sensitive to seed count, aggregation, and interval construction [21]. Tracking adds a conditioning problem: kinematic error among survivors can improve when a method terminates earlier. HumanTracker similarly finds that kinematic metrics can miss support and contact failures [22]. We therefore use a liveness-weighted primary score and report common-survivor quality only beside survival.

III. CLIMB: Feasibility-Gated Tracking

A. Problem and assumptions

Let a retargeted reference provide generalized position and velocity $(q_t, \dot{q}_t)$ for the G1 at frame t. We apply a centered five-frame moving average to velocity, differentiate at the reference frame rate, and smooth acceleration with the same window. Feasibility is defined relative to a robot model, scene, collision geometry, friction, actuator ranges, and reference. The present instantiation uses the G1 and a flat plane; electrical, thermal, compliance, latency, and structural limits remain outside this model-relative label.

We use the compiled MuJoCo model underlying the tracking environment. Contact-free inverse dynamics returns the generalized force required for the prescribed acceleration. The six free-joint coordinates define the base wrench W_t that the environment must supply; the remaining coordinates define contact-free joint torque τ_t^{free}. A collision geometry becomes a candidate contact when its exact distance to the plane is at most $g = 0.06\,\text{m}$.

The 6 cm band is a declared tolerance. At 3 cm a known feasible control is flagged on 42% of frames because the bank carries an approximately 3 cm ground-alignment offset. At 10 cm the attractor's 7–10 cm floating descent is incorrectly granted a contact candidate. The chosen setting lies between these observed failure modes; it is not a universal constant.

B. Contact-capacity screen

For each candidate point we construct a four-edge pyramidal friction cone with coefficient 0.6 in the primary model; a simulator-matched view makes non-foot contacts frictionless. Mapping nonnegative generator magnitudes f_t through the contact Jacobian yields both base-wrench matrix $A_t f_t$ and joint-torque contribution $J_t f_t$. A weighted nonnegative least-squares problem first exposes the unconstrained residual, with angular-wrench rows weighted by two. We then solve a linear program minimizing ℓ_1 wrench slack subject to

$$-\tau_{\max} \le \tau_t^{\text{free}} - J_t f_t \le \tau_{\max}, \qquad f_t \ge 0. \tag{1}$$

The translational component of the resulting weighted residual defines unsupported force. With no candidate contact, the residual is the contact-free demand.

We report the fraction of frames with no contact candidate (`airborne_frac`), the fraction whose torque-limited unsupported force exceeds half the robot weight (`infeasible_frac`), normalized unsupported-force integral, and joint-torque infeasibility. A clip crosses the primary rule only when `infeasible_frac`> 0.10 (strict inequality). Ballistic flight is not flagged merely for being airborne: near-gravity acceleration has little external support demand. Conversely, nonballistic hovering or descent without support is flagged.

The screen takes approximately one CPU-second per clip in the primary implementation. An independent implementation processes the production bank at 0.145 CPU-s per clip (0.84 ms/frame). The screen therefore runs as an offline bank-ingestion stage rather than in the policy control loop.

C. Routing and exact-support curriculum

Clip classification is too coarse because one reference can contain both supportable and unsupported intervals. We reduce per-frame screens to ordered feasible runs, retain only 50-step windows wholly inside a run, and bind each motion and sidecar by SHA-256. Across 800 training clips, 1,841 source runs yield 1,184 admissible units and 368,951 legal starts after 657 short runs are discarded. A unit is an attribution stratum; the deployment prior is uniform over legal starts, not units or clips.

Let $\mathcal{U}$ contain candidate units, $F_u \in \{0, 1\}$ denote admission under the declared screen, and n_u count legal starts. The duration-correct base mass is $b_u = n_u / \sum_v n_v$. For absolute learning progress $\mathrm{LP}_u(k)$ at sampler clock k, CLIMB forms

$$q_u(k) = \frac{F_u b_u [\mathrm{LP}_u(k) + \lambda]}{\sum_v F_v b_v [\mathrm{LP}_v(k) + \lambda]},$$
$$p(k) = \mathrm{Cap}_{c_{\mathrm{unit}}, c_{\mathrm{clip}}} \left(\bar{b}, q(k); \rho\right), \tag{2}$$

where $\bar{b}$ is b normalized over admitted units. Without an active concentration limit, the operator returns $\rho\bar{b} + (1-\rho)q$; otherwise it reallocates only the focused mass under the fixed unit and clip ceilings while preserving the exact lower bound $p_u \ge \rho\bar{b}_u$. Thus a rejected interval receives zero mass under the declared model, while ρ preserves deployment-prior exploration inside the gate. The present method uses a binary F_u so admissibility and allocation remain separately testable; a smooth residual-based gate would be a different intervention.

Intervals do not all receive the same remedy. Admissible runs enter the exact support. References whose required contact object is absent from the scene are paired with that context or withheld. A lightweight root-translation plus contact-IK repair path can project eligible cases before the same screen and qualification checks are rerun; residual infeasibility, excess displacement, joint-limit failure, or contact-IK residual sends the result to quarantine. This routing preserves the original motion rather than treating every screen flag as a deletion instruction.

The runtime samples only legal starts, emits explicit truncation at a segment boundary, never wraps into a rejected frame, and serializes its deterministic generator and sufficient statistics. Per-unit and per-clip caps bound concentration. Missing sidecars, changed motion hashes, invalid starts, invalid frames, or censored resets fail the contract.

The paired evaluator freezes the motion, start, replicate, environment-noise seed, and dynamics-noise seed across policies. It assigns the reference before the first observation, disables auto-reset until terminal channels are captured, rejects invalid offsets, and records checkpoint, task, condition, reference, evaluator, and output hashes. The 100-clip panel is name- and hash-disjoint from training and contains 2,800 paired conditions.

IV. Experimental Design

A. E1: RPM attractor diagnosis

We analyze the previously sealed 100-motion campaign: 4,096 environments, 4,000 PPO iterations, three seeds per arm, and 100 disjoint evaluation motions with eight episodes per clip. It includes failure-adaptive, clip-uniform, and normalized grounded samplers. We report exposure, attractor identity, held-out survival, and the attractor's modeled contact-capacity residual. This campaign establishes the motivating failure case; E4 separately isolates the exact-support allocator.

B. E2: bank scale and implementation agreement

The primary corpus contains 10,705 AMASS-derived references retargeted through `whole_body_tracking` to G1. We report raw counts under the fixed rule, category/source breakdowns, and contamination in the historical evaluator. A separate experiment covers all 4,950 references in the filtered BONES-SEED production pairing. We never pool their prevalence. A deterministic 40-clip panel, 20 from each bank and enriched for flags, passes through both implementations to measure score and decision agreement; it is not a prevalence sample.

C. E3: filter, repair, and allocation controls

We test whether simpler routing choices already establish training benefit. E-HYG prunes 99 flagged motions from an 800-motion bank but yields a held-out feasible-motion effect of -0.0101 under its predeclared test. Soft segment sampling estimates -0.0196 with a 95% hierarchical-bootstrap interval $[-0.0497, +0.0134]$, but fails its rejected-start-mass manipulation gate. An exploratory segment-native pilot reaches only 0.014 total variation (TV) from control. These negative results motivate an exact-support, manipulation-first test.

We also exercise the repair route without treating repair as a policy win. A source- and severity-stratified CPU panel applies root translation plus contact IK, reruns the feasibility screen, and requires residual infeasibility at most 5%, root displacement at most 8 cm, joint-limit compliance, and contact-IK residual at most 10 mm. It admits 22/26 flagged candidates and keeps 4/4 feasible controls byte-identical, producing 36 units and 10,561 legal starts. A separate sealed repair-all policy study is reported as a boundary because it misses its benefit and coverage gates.

D. E4: feasibility-gated allocation ablation

Phase G compares two 512-environment, 4,000-iteration arms with confirmation seeds 1–3. G1 samples uniformly over legal starts. G2 estimates each unit's Beta-smoothed conditional success rate $s_u(k)$ and ranks absolute learning progress

$$\mathrm{LP}_u(k) = |s_u(k) - s_u(k-10)|. \tag{3}$$

Its focus distribution is proportional to the legal-start prior times $\mathrm{LP}_u + \lambda$; exploration ρ mixes the G1 prior back in before fixed unit and clip caps.

Before sealing, a 12-row grid crosses $\rho \in \{0.05, 0.10, 0.20, 0.40\}$ and $\lambda \in \{0.001, 0.01, 0.05\}$. Each row runs 50 iterations on seed 20260903. Only sampler ledgers at iterations 30, 40, and 49 enter selection. A row passes when mean TV is in $[0.05, 0.15]$, each checkpoint lies in $[0.025, 0.20]$, entropy-effective units are at least 12,

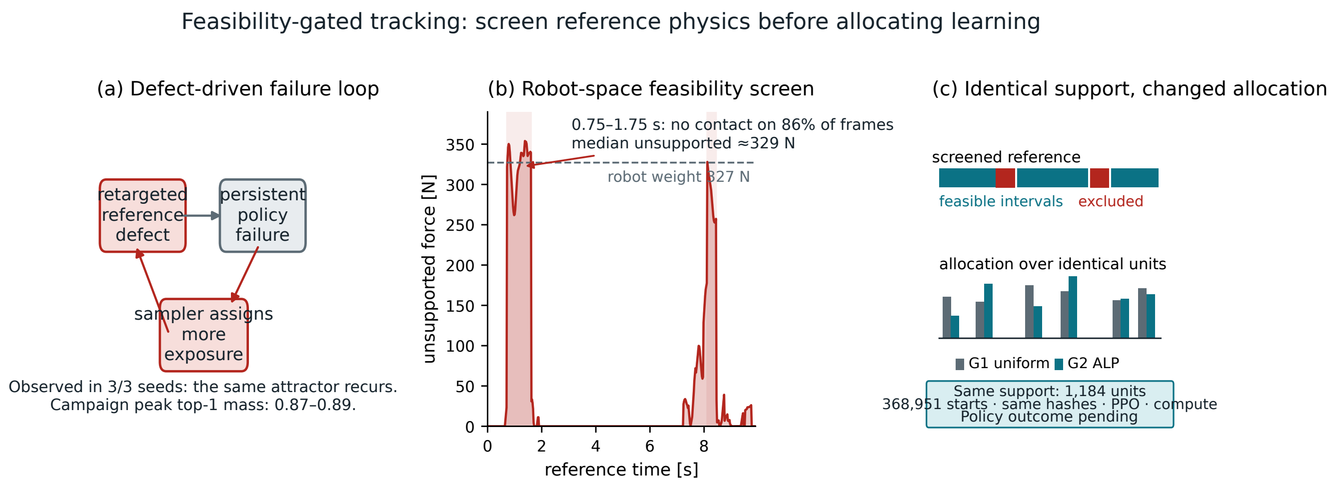

Fig. 1. **The CLIMB feasibility-gated tracking interface.** (a) In the motivating simulation campaign, the same attractor recurs in 3/3 adaptive seeds; 0.87–0.89 is the campaign maximum top-1 mass and does not belong to that clip in every seed. (b) A reference-only screen localizes the unsupported interval. This pipeline-specific temporal and modeled-wrench diagnosis isolates this failure case rather than every cause of adaptive concentration. (c) The preregistered G1/G2 comparison keeps 1,184 units, 368,951 legal starts, hashes, PPO, and compute fixed while changing allocation; its policy outcome is pending.

top-1 probability is at most 0.05, invalid/censored counts are zero, and final rank saturation is below 0.90. The passing row nearest TV 0.10 is repeated on seed 20260904. Failure returns the study to design without endpoint access.

After a 400-iteration warm-up, the confirmation gate applies the same separation, concentration, validity, seed, and provenance checks. Only a passed gate unlocks the primary endpoint: liveness-weighted TrackingScore on 25 reference-defined feasible-hard clips,

$$h \exp\left(-\frac{\text{MPKPE}}{0.30\,\text{m}} - \frac{\text{anchor error}}{0.40\,\text{rad}}\right), \tag{4}$$

where h is survived-horizon fraction. The independent unit is clip within training seed, with a seed-then-clip hierarchical bootstrap of 10,000 draws. Survival, all-panel TrackingScore, AULC, and common-survivor non-harm are secondaries. Exactly one status is printed: positive, null, inconclusive, or `not_tested`.

V. Results

A. RPM creates a curriculum attractor

Adaptive peak top-1 mass is 0.884/0.870/0.893 across campaign seeds, and the same BMLmovi kneel-and-crawl clip repeatedly becomes dominant in all three. The maximum belongs to another clip in two seeds. Uniform sampling achieves higher held-out endpoint survival in every paired seed: differences (uniform minus adaptive) are +0.0300, +0.0275, and +0.0300. With only three seeds, the sign pattern is the useful description; the minimum attainable paired permutation p-value is 0.125.

The reference has no joint-limit violation and peak joint speed is at most 5.6 rad/s. The dynamic screen instead localizes the failure (Fig. 1). From 0.75–1.75 s, 86% of frames have no candidate contact and median unsupported force is 329 N; the model weighs 327 N. The 1.75–7.25 s kneeling interval is supportable through non-foot contacts, and the 8.0–8.5 s rise again becomes unsupported. Every trained policy has zero survival from frame zero, while the tested late 8 s offset survives.

B. The gate is pipeline-conditioned and reproducible

Under the strict rule, 2,442 of 10,705 primary-pipeline clips are flagged (22.8%). Category rates range from 12.9% for quiet motion to 58.6% for dynamic motion, and source rates from 0.1% for GRAB to 100% for CNRS. Severity-stratified inspection shows why labels are insufficient: five inspected CNRS cases are ordinary fast walks with extended floating intervals, while the Transitions sample mixes ingest defects with acrobatic content.

Only 7 of 4,950 production clips (0.14%) cross the same nominal rule, while 111 (2.24%) are airborne on more than 10% of frames. Five of seven flags are jumps, four naming a 50 cm box absent from the flat audit scene. Their verdict is *unsupported on this plane*, routing them toward missing terrain or exclusion rather than geometric root projection.

Across the stratified 40-clip panel, `infeasible_frac` rank correlation is 0.9836, `airborne_frac` correlation is 0.9974, and strict decisions agree for 39/40 clips (Cohen's $\kappa = 0.9485$). The disagreement is `burpee_002__A362_M` (0.019 versus 0.136). Figure 2 keeps the full-bank denominators separate and uses only this same-input panel to check implementation agreement; it does not establish corpus or retargeter equivalence.

C. Routing changes support, but policy benefit is conditional

The historical evaluator contains 29 flagged clips among 100. Across three policies, flagged clips score 6.0–8.4 survival points below the all-clip aggregate and 8.4–11.8 below the feasible stratum. The grounded-minus-uniform endpoint contrast is +0.025 on feasible clips and −0.009 on flagged

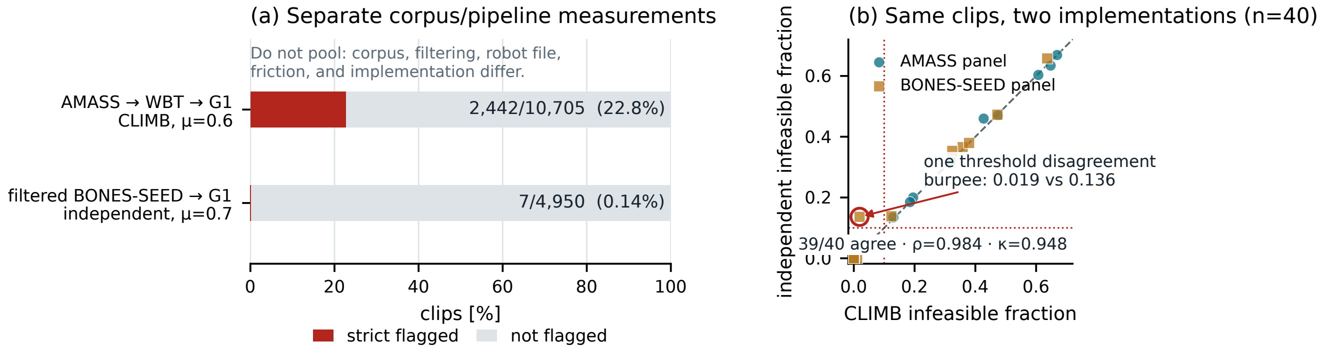

Fig. 2. Bank-scale screen and implementation agreement. (a) Bars are normalized within two separate corpus/pipeline denominators and apply the strict `infeasible_frac`> 0.10 rule: 2,442/10,705 for AMASS→WBT→G1 and 7/4,950 for filtered BONES-SEED→G1. Corpus, filtering, robot file, friction, and implementation differ, so this is not a causal retargeter comparison. (b) On a flag-enriched, deterministic 20+20 same-clip panel, strict decisions agree for 39/40 clips (Spearman ρ = 0.9836, Cohen's κ = 0.9485). That panel checks implementation agreement; its stratified selection is not a prevalence sample.

clips. These data justify stratified evaluation, but changing the evaluation subset cannot establish training benefit.

The routing controls separate operator validity from policy value. E-HYG's pruning contrast is a sealed null, soft segment sampling does not pass its allocation gate, and the prior segment-native treatment is only 0.014 TV from control. Together they distinguish three statements often collapsed in discussion: a defect can be measured; screening changes admissible support; and an allocator can improve policy performance on that support. Phase G is the direct test of the third.

The repair panel demonstrates that the routing operator can recover exact-support candidates: 22/26 flagged cases pass every residual, displacement, joint-limit, contact-IK, and hash gate, while 4/4 feasible controls remain byte-identical. The panel is stratified rather than a prevalence sample. In the earlier repair-all policy study, repaired-reference deployment changes by +0.0397 with a motion-bootstrap 95% interval $[+0.0153, +0.0658]$, below its +0.05 smallest effect of interest; unchanged-reference policy transfer is −0.0036, and the coverage gate fails. Repair is therefore an implemented route whose downstream value requires a matched, distortion-aware comparison, not the explanation for the present policy claim.

D. Feasibility-gated allocation on exact support

Phase-G status: endpoint-blind calibration in progress. The exact 900-motion payload passes all committed identities. The 12-candidate, endpoint-blind calibration is queued behind a 14,000 MiB shared-GPU availability gate. This subsection will be populated only from the frozen result table. If manipulation or provenance fails, the paper will report `not_tested` and no endpoint estimate. If all gates pass, it will report the feasible-hard TrackingScore contrast with seed-then-clip interval, survival, AULC, and common-survivor non-harm.

VI. Limitations

CLIMB's admission label is model-relative. The current screen instantiates one G1 model and a flat scene with collision contacts, friction, and actuator force limits. Hardware deployment would add electrical, thermal, compliance, latency, and structural constraints and then require closed-loop validation. Terrain-bearing references likewise require their terrain in the scene; the four box-jump flags in the production bank are routed to missing context rather than declared bad motion.

Finite differences, the 6 cm contact band, and the half-weight residual threshold introduce modeling choices. Local sensitivity checks bound the two thresholds on this bank, but a new robot or retargeter must recalibrate them. The 22.8% and 0.14% rates therefore remain separate corpus–pipeline measurements. A smooth feasibility weight derived from residual slack could retain borderline cases, but would couple admissibility and allocation; it requires its own matched ablation against the current binary gate.

Phase G has three training seeds, or two under the predeclared budget fallback, so uncertainty is seed- and clip-structured. Its result applies to the calibrated ALP rule, support, task, and budget. The repair route likewise needs a same-policy, distortion-aware comparison of certified repair against exact feasible intervals and quarantine. The observation that would settle this is joint movement of tracking, survival, reference fidelity, contact timing, and mechanical work under one paired trial contract.

VII. Conclusion

Reference–physics misalignment turns persistent tracking error into a misleading curriculum signal. In the diagnosed three-seed campaign, one such reference repeatedly becomes a dominant attractor; the screen localizes its unsupported wrench demand before policy training. CLIMB turns that diagnosis into an active interface: it screens and routes final robot-space trajectories, constructs exact legal-start support, and assigns zero training mass to rejected intervals under the declared model. The pending matched allocation ablation will determine whether ALP adds value inside that gate. In either outcome, feasibility and learning progress are now separate quantities that can be measured, changed, and evaluated independently.

Reproducibility and AI disclosure

The screening and training stack uses mjlab v1.6.0 at commit `0fb8a681136b`, MuJoCo 3.11.0, NumPy 2.5.1, and SciPy 1.16.2. Motion, sidecar, unit-table, task, checkpoint, evaluator, condition, and output identities are SHA-256 bound. Released code and aggregate artifacts will be supplied through an anonymous review repository; licensed AMASS-derived motion files cannot be redistributed. OpenAI Codex assisted with code, figure composition, and language editing. The authors verified scientific claims and artifact provenance.

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